Physica A 675 (2025) 130836



Contents lists available at ScienceDirect

Physica A: Statistical Mechanics and its Applications

journal homepage: www.elsevier.com/locate/physa





Cascading failures in interdependent transportation networks: A generalized coupled map lattice model with intermodal dynamics

Biao Chen [a], Jianjun Wu [b], Yanyan Chen [a], Der-Horng Lee [c], Zhengbing He [d,*]

[a] Beijing Key Laboratory of Traffic Engineering, Beijing University of Technology, Beijing, China
[b] School of Economics and Management, Dalian University of Technology, Dalian, China
[c] ZJU-UIUC Institute, Zhejiang University, Haining, China
[d] Laboratory for Information and Decision Systems (LIDS), Massachusetts Institute of Technology, Cambridge, United States

ARTICLE INFO

Keywords:
Multimodal transportation network
Cascading failure
Coupled map lattice
Intermodal coupling
Generalizability

ABSTRACT

Urban public transportation networks constitute critical infrastructure essential for daily mobility. During emergencies, the failure of stations or lines may trigger cascading failures. In multimodal transportation networks, disturbances influence both intramodal and intermodal operations. While existing research has examined the cascading failure in multimodal networks by considering network topology and travel demand, the inherently complex interdependencies within such networks remain a challenge. To address this, this study improves the weighted coupled map lattice model to analyze failure propagation. The improved model integrates network structure and travel demand to determine inter-node weights, effectively capturing coupling effects between transportation modes. An interdependence factor that integrates both spatial and functional interdependencies is introduced into the node state evolution function. Spatial interdependence reflects static intermodal connections based on the distance between coupled nodes, while functional interdependence represents dynamic interactions measured by variations in betweenness centrality and demand following disturbances. To enhance model generalizability, two discrimination parameters are introduced. Finally, the proposed model is applied to a bus-subway network in Tianjin, using smart card transaction data to construct a weighted multimodal transportation network. Results reveal that the improved CML model effectively captures the intermodal and intramodal node dynamics. The residual capacity plays a significant role in enhancing the network robustness. Furthermore, an appropriate coupling distance should be established in a multimodal network, which can provide convenience for passengers and reduce the losses during cascading failures. This study offers a theoretical foundation for transportation managers to develop emergency strategies and mitigate disruptions.

1. Introduction

With the growing travel demand and the diversification of transportation modes, urban multimodal transportation networks have emerged to satisfy these needs. However, the systems are vulnerable to disturbances caused by emergencies such as extreme weather, equipment failures, or terrorist attacks [1,2]. These disturbances can severely impact daily mobility and may lead to substantial economic losses. For example, in August 2003, the North American blackout triggered a cascading failure that paralyzed

* Corresponding author.
E-mail address: he.zb@hotmail.com (Z. He).

https://doi.org/10.1016/j.physa.2025.130836
Received 20 March 2025; Received in revised form 17 July 2025; Accepted 19 July 2025
Available online 20 July 2025
0378-4371/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

transportation networks, including roadways, airports, and subways (with 600 trains stranded) across the region. This event caused widespread passenger disruptions and substantial economic losses. Similarly, in July 2022, air traffic control measures led to the cancellation of hundreds of flights and affected 12 airports, which caused extensive delays for over 20 days. These examples illustrate that disturbances in transportation infrastructure not only affect normal operations but also trigger cascading failures. Moreover, in large cities, daily commuting patterns exhibit a bimodal distribution [3], i.e., morning and evening peak hours. The rapid surge in passenger flow within a short period can overload transportation facilities, which increases the risk of failure. Under a disturbance, uncertainties in travel demand and rerouting strategies yield complex ripple effects within the network, particularly in multimodal transportation networks. Therefore, it is essential to develop a model to simulate the propagation of disturbance impacts in multimodal transportation networks and capture the dynamics of nodes during emergencies to improve the resilience and operational efficiency of urban transportation systems.

In network science, cascading failure incorporates both the interactions and state evolution of network components, which offer a more comprehensive framework for quantifying continuous state transitions. Cascading failure processes primarily investigate chain reactions in which the failure of one or more system components triggers subsequent failures in interconnected components, which potentially leads to large-scale collapse [4]. In urban transportation networks, disturbances, such as equipment malfunctions, operational failures, or sudden demand surges, can impact normal operations. These disturbances can lead to the closure of one or more stations and force passengers to reroute through adjacent stations or opt for alternative transportation modes. Consequently, this redistribution of passenger flow may cause congestion or operational failures at other stations, potentially escalating into widespread network paralysis [5,6]. This phenomenon exemplifies a cascading failure process so that it can be effectively analyzed using cascade failure models. Therefore, investigating cascading failures in transportation networks is crucial for understanding the dynamic propagation of disturbances and evaluating network performance under severe disturbances.

The increasing diversification of urban transportation modes has strengthened coupling effects within interconnected multimodal transportation systems. Recent studies model these interdependent networks by establishing “one-to-one” or “one-to-many” intermodal links based on Euclidean distance between nodes to represent coupling strength or transfer probabilities. However, intermodal transfers inherently depend not only on spatial proximity but also on travel demand patterns, which indicate that distance-based coupling relationships may neglect critical nonlinear interactions in multimodal mobility networks. Moreover, existing research on cascade dynamics predominantly focuses on static features, such as topological properties or passenger flows, while neglecting dynamic node properties. Additionally, existing failure propagation models are often constrained to idealized monomodal or multimodal networks, which limit their applicability to diverse transportation networks.

To address these limitations, this study integrates transit transaction data to construct a weighted multimodal transportation network and explores an improved Coupled Map Lattice (CML) model to capture the cascading failures in such networks. The improved CML model enhances its ability to capture intermodal propagation dynamics and increase its generalizability. Finally, the effectiveness of the proposed model is validated using a case study of multimodal networks in the six central districts of Tianjin. The main contributions of this study are as follows:

- A method for determining coupling weights, which considers both coupling distance and multimodal travel demand, is proposed. This study establishes a more comprehensive intermodal coupling relationship by integrating coupling distance and multimodal travel demand. Moreover, network edge weights are determined by integrating network topology and travel demand, which enables a more precise depiction of disturbance propagation between nodes.
- An integrated interdependence factor, combining spatial and functional interdependence, is developed and incorporated into the CML model to reflect the static and dynamic properties of the node states during cascading failures. This enhancement strengthens the role of intermodal coupling nodes in the analysis of cascading failures in multimodal transportation networks.
- Two discrimination parameters are introduced in the CML model to determine the type of demand and the degree of coupling strength. These modifications extend the applicability of the model to cascading failures in both monomodal and multimodal transportation networks, which enhances its generalizability.

The structure of this paper is organized as follows. Section 2 reviews the literature on related research. Section 3 introduces the methodology for constructing multimodal networks, presents the development of a disturbance propagation model for multimodal networks, and proposes key indicators for evaluating network performance during cascading failures. Section 4 validates the proposed models using a real-world network case study. Finally, Section 5 provides the concluding remarks.

2. Related work

Cascade failure is an extreme phenomenon in the study of network disturbance impact propagation, which is primarily analyzed in complex systems with network characteristics, such as power, communication, and transportation networks [7]. Recently, several scholars have examined the propagation of disturbances in urban transportation networks using epidemic dynamics models, such as the Susceptible-Infected-Recovered (SIR) and Susceptible-Infected-Susceptible (SIS) models [8,9]. Moreover, Li et al. [10] proposed a speed perturbation model for connected and automated vehicles to investigate the disturbance propagation in road transportation systems.

In the context of transportation networks, numerous studies have demonstrated the effectiveness of cascading failure models in capturing cascading effects and state evolution processes under disturbances [11,12]. For example, Jin et al. [13] developed a simulation-based model for rail transit systems that simultaneously accounts for demand variations and time delays to analyze

cascading failure dynamics. Moreover, several studies have highlighted that the structural features of transportation networks significantly influence the cascading failure process. In particular, transportation networks with scale-free properties exhibit greater vulnerability to deliberate disturbances [14,15]. Furthermore, researchers have explored the impact of node failure duration on the cascading process [16], the spatial distribution of failed nodes [17], and corresponding mitigation strategies [12].

In recent years, researchers have developed various mathematical models to simulate cascading failure in complex networks, including the Sandpile model [18], the OPA model [19], the "load-capacity" model [17,20–23], and the CML model [24–26]. Among these, the Sandpile and OPA models are widely applied to energy, power, and information networks. The "load-capacity" model is frequently employed to analyze cascading failures in transportation networks due to its alignment with travel demand and network capacity in transportation networks. For example, Duan et al. [27] incorporated travel demand, capacity, and passenger route choices into a dual-layer cascading failure model to evaluate transportation network reliability. Similarly, Shen et al. [28] focused on a single subway station and categorized passengers within the station into three groups. They constructed facility service chains tailored to each passenger group and developed a hierarchical network based on service chains to analyze cascading failures by integrating a "load-capacity" model. However, the "load-capacity" model assumes a positive correlation between the maximum load capacity of a station and its initial load, whereas Kim and Motter [29] revealed that demand and load do not always exhibit a strictly linear relationship. With the expansion of urban transportation networks and the increase in travel demand, the traditional "load-capacity" model has shown significant limitations in analyzing cascading failures. To improve analytical accuracy, researchers have developed nonlinear "load-capacity" models. For instance, Ma et al. [30] applied a nonlinear "load-capacity" model to assess the multimodal transportation network robustness from structural and functional perspectives. Additionally, Guo et al. [31] incorporated failure recovery functions to analyze the impact of the initial load, failure and recovery mechanisms, and recovery speed on cascading failures in multimodal transportation networks.

The CML model is widely used to analyze cascading failures in chaotic systems, especially complex networks [32]. In recent years, it has been extensively applied to the study of cascading failures in transportation networks [33–36]. For example, Xu et al. [26] investigated the cascading failure processes in the multimodal transit network using a topology-based CML model, thereby evaluating the reliability of such networks under disturbances. However, in addition to network structure, travel demand significantly influences transportation network performance. Recently, researchers have improved the model by integrating travel demand into the topological structure to better capture the state evolution of transportation networks. For example, Sun et al. [24] proposed an improved CML model considering both network topology and passenger flow to analyze the vulnerability of rail transit networks under cascading failures. Most studies utilizing the CML model have primarily focused on subway networks [33,34,37]. However, with the diversification of urban transportation modes, researchers have extended the model to more accurately represent cascading failures in multimodal networks. For instance, Zhang et al. [33] introduced relative departure frequency that is equivalent to travel demand and incorporated geographic and operational coupling factors into the CML model to examine cascading failures in multimodal transportation networks. Their study utilized an R-space representation of the bus network to capture cascading failures at the route level. Moreover, the relative departure frequency and geographic coupling in the model are static attributes, which limit the ability of the model to reflect dynamic changes in network topology following node failures over time. Building on this approach, this study uses subway stations and bus stops as nodes and fully accounts for the impact of dynamic structural and demand changes during cascading failures, thereby enhancing the accuracy of the CML model for analyzing cascading failures in multimodal transportation networks.

To quantify the impact of cascading failures on transportation network performance, many studies examine variations in network performance from structural and functional perspectives. Existing studies commonly assess network robustness [25,30], vulnerability [7,37,38], and reliability [27,39] by employing metrics such as the proportion of failed nodes, the maximal connected subgraph ratio, network efficiency, and passenger loss rate. For example, Gao et al. [35] introduced metrics including the maximal connected subgraph and network efficiency to evaluate a two-layer rail transit network, and analyzed its dynamic robustness using an extended CML model. Zhao et al. [40] analyzed the resilience of four expansion plans using the curve of the maximum connected subgraph ratio changing over time. Furthermore, critical node identification is a key aspect of studying network performance under failure scenarios. For example, Li et al. [41] proposed a dynamic station-line centrality to identify critical stations within bus-metro networks. Accurately identifying critical nodes not only enhances network robustness and suppresses cascade failure propagation but also provides support for transportation managers to prioritize repair strategies. Therefore, analyzing critical nodes during cascade failure processes holds significant importance.

Compared to cascading failures in monomodal transportation networks, those in multimodal networks involve both intramodal and intermodal propagation. Accurately quantifying interdependencies between networks is essential for understanding node state evolution and demand redistribution processes [42]. Most existing studies on cascading failures in multimodal networks rely primarily on spatial distance to establish coupling relationships and characterize node state evolution based on network topology [26,43]. However, when stations or stops fail, passenger transfer behavior is influenced by multiple factors beyond spatial proximity. To improve the accuracy of cascading failure modeling, Zhang et al. [33] incorporated both spatial distance and relative departure frequency into the state evolution function of the CML model to reflect network interdependencies. Additionally, variables such as bus lines, transfer time, and transfer costs also significantly affect passenger transfer behavior [44,45]. Based on available data, this study integrates both spatial distance and multimodal travel demand to characterize inter-network coupling relationships.

In summary, epidemic models are widely utilized to represent discrete state transitions of network components (i.e., nodes or edges) and offer a macroscopic description of network state variations, integrating statistical methods for comprehensive analysis. However, these models often depend on predefined transmission rates and are constrained to the fixed discrete states of network components, limiting their ability to capture continuous state changes. Meanwhile, the "load-capacity" model captures the disturbance propagation by relating traffic flow to node load in transportation networks, which offers a rapid, concise, and intuitive framework for

modeling cascading processes. Although this model has been progressively refined in recent years, it still exhibits limitations in simulating interaction between nodes and capturing the nonlinear evolution of node states. In contrast, the CML model is capable of describing the coupled evolution of nodes with discrete-time dynamics and continuous states in networked systems. It can effectively capture cascading failures in networks, considering network structure, network capacity, and the relationship between load and capacity simultaneously, and simulate nonlinear coupled systems under continuous-time scenarios. However, it is highly sensitive to initial conditions and parameter settings, accompanied by relatively high computational complexity. Building on these insights, this study proposes an improved CML model to investigate cascading failures in multimodal transportation networks.

Furthermore, interactions among various transportation modes reinforce their interdependencies within multimodal transportation networks. Analyzing a single transportation network in isolation fails to fully capture the impact of disturbance propagation across the entire transportation system. However, existing studies predominantly focus on monomodal transportation networks. Most CML models, which are often used to simulate cascading failures in multimodal transportation networks, rely on network topology to represent intermodal coupling relationships and overlook the crucial role of multimodal travel demand in intermodal coupling effects. Integrating both coupling distance and multimodal travel demand into the disturbance propagation process can significantly enhance model accuracy. Furthermore, current research on cascading failures in multimodal transportation networks primarily employs static attributes to represent intermodal coupling, which limits the ability to capture dynamic coupling characteristics during cascading failures. To address these limitations, this study constructs a multimodal transportation network that considers both topological structure and travel demand and proposes an improved CML model that introduces spatial and functional interdependence to investigate the cascading failure in a multimodal transportation network.

3. Methodology

3.1. Multimodal transportation network model

This study examines two transportation modes (subway and bus networks) within an urban public transit network. The M-space representation is utilized to construct a multimodal transportation topology network [46]. Intermodal coupling relationships and weights are determined by integrating distances between nodes with multimodal travel demand, ultimately forming a weighted multimodal transportation network model. The process involves the following steps: First, the L-space representation is used to construct separate subway and bus networks. Specifically, subway stations and bus stops are treated as network nodes, and connections are defined by adjacency relationships between stations or stops within subway or bus lines. Subsequently, intermodal coupling relationships are established. To enhance the accuracy of these relationships, this study introduces a coupling rule that incorporates transfer distance and multimodal demand, aiming to capture the intramodal and intermodal propagation of disturbances within multimodal networks. Finally, multiple transportation modes collectively form a complex and interconnected multimodal transportation network, as illustrated in Fig. 1. The notations used in the problem formulation are summarized in Table 1.

3.2. Network interdependence factors

(1) Spatial interdependence

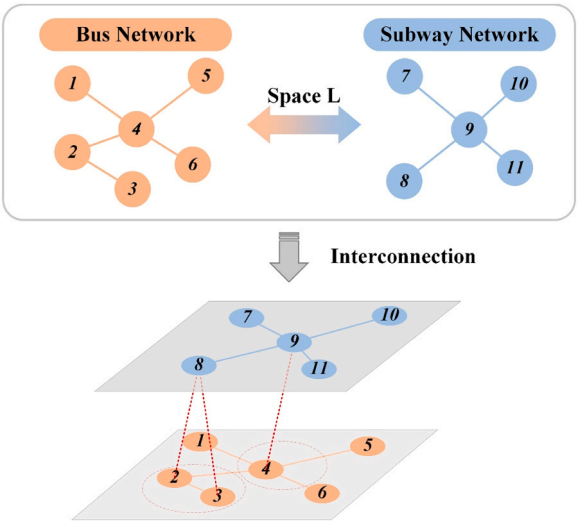


Fig. 1. Schematic diagram of multimodal transportation network construction.

Table 1
Notations used for problem formulations.

Notations	Description
Dis_{ij}	The coupling distance between intermodal nodes i and j
spa_{ij}	The nonlinear spatial interdependence strength between intermodal nodes i and j
SI_i	The spatial interdependence strength of coupling node i
N_{Ci}	The set of nodes connects with the coupling node i
N_C	The set of the coupling nodes
BC_{ij}^t	The betweenness centrality of the coupling edge between node i and node j at the t-th time step following a disturbance
o	The origin of passenger travel
d	The destination of passenger travel
N	The set of nodes in the multimodal network, $\|N\|$ denotes the number of nodes in this set
p_{od}	The number of shortest paths between o and d
p_{od}^{ij}	The number of paths that pass through the nodes i and j between nodes o and d
FI_i^t	The functional interdependence factor of node i at the t-th time step
IC_i	The integrated interdependence factor of node i, and when node i is a non-coupled node, $IC_i = 0$
q_i	The travel demand of node i at the initial state
C_i	The maximum load capacity of node i
$x_i(t)$	The state value of node i at the t-th time step, $x_i^{initial}$ denotes the initial state of node i
$f(x)$	The chaotic system mapping function, which has been widely used in studies to characterize the evolution patterns of capacity-constrained transportation systems
W_{ij}	The comprehensive weight of the link between nodes i and j
$S_i(t)$	The sum of the weights of the links connected to node i at the t-th time step
w_{ij}^T	The topological weight between nodes i and j
w_{ij}^D	The flow weight between nodes i and j
a_{ij}	The connectivity between nodes i and j, if nodes i and j are connected; $a_{ij} = 1$; otherwise, $a_{ij} = 0$
q_{od}	The travel demand between o and d
f_{ij}	The flow between nodes i and j, obtained from transit transaction data
$f_{jk}(t+1)$	The passenger flow on the link between node j and node k at time step $t+1$
Δf_{jk}	The increment in passenger flow on the link between node j and node k
R	The external disturbances
$c(t)$	The disturbance propagation scale at the t-th time step
$N_d(t)$	The set of failed nodes in the network at the t-th time step, and $\|N_d(t)\|$ indicates the number of failed nodes in this set
$MCS(t)$	The network connectivity indicator that measured by the size ratio of the maximum connected subgraph
$N_{max}(t)$	The nodes of the maximum connected subgraph at t-th time step, $\|N_{max}(t)\|$ denotes the number of nodes of this set
$p(t)$	The proportion of newly failed nodes of mode k at the t-th time step
$N_d^{new}(t)$	The newly failed nodes at the t-th time step, $\|N_d^{new}(t)\|$ is the number of nodes in this set
$N_d^{M1}(t)$	The newly failed nodes caused by the intermodal propagation pattern at the t-th time step
$N_d^{M2}(t)$	The newly failed nodes caused by the intramodal propagation pattern at the t-th time step
α	The adjustment coefficient for topological and flow weights. To balance these two weights, $\alpha = 0.5$
ζ	The adjustment coefficient for the spatial and functional interdependence factors
ε	The node correlation coefficient. It quantifies the degree to which the state of node i is influenced by its adjacent nodes
γ	The adjustment coefficient for the coupling strength of cascading effects between modes
τ_{ij}	The demand-type discrimination parameter
δ_{ij}	A binary variable, if nodes i and j are located in networks of different transportation modes and have a coupling relationship, $\delta_{ij} = 1$, otherwise, $\delta_{ij} = 0$
ρ_{ij}	A binary variable, if the path between o and d passes through nodes i and j, $\rho_{ij} = 1$, otherwise, $\rho_{ij} = 0$

The spatial interdependence of multimodal transportation networks is established by selecting a primary transportation mode and identifying corresponding nodes from secondary networks within a specified coupling radius around each node in the primary network. However, multiple coupling nodes may be determined within this radius, each at different distances. To accurately capture these variations, this study introduces a distance-dependent interdependency function from geographic dependency research to quantify the coupling effects between nodes [47]. This function models the nonlinear relationship between infrastructure correlation and spatial distance, which indicates that the correlation between facilities diminishes with increasing distance. This pattern is consistent with passenger transfer behavior in transportation networks. Building on this function, the spatial interdependence relationship (Spatial Interdependence, SI) between transportation modes is quantified based on the geographical locations of bus and subway network nodes. The formula is expressed as follows:

$$SI_i = \frac{\sum_{j=1}^{N_{Ci}} spa_{ij} \cdot \delta_{ij}}{\sum_{i=1}^{N_C} \sum_{j=1}^{N_{Ci}} spa_{ij}} \tag{1}$$

$$spa_{ij} = 1 - \log_{11}(1 + Dis_{ij}) \tag{2}$$

where $spa_{ij} \in [0,1]$. In this study, the coupling distance is defined within the range of 0–1000 m (i.e., $Dis_{ij} \in [0,10]$, in units of hundred meters). Accordingly, the base of the logarithm is set to 11, which results in spa_{ij} ranges from $1\text{-}\log_{11}(1+10) = 0$–$1\text{-}\log_{11}(1+0) = 1$.

(2) Functional interdependence

Compared to monomodal transportation networks, multimodal transportation networks exhibit greater redundancy to disturbances, as passengers have access to more alternative routes after disturbances. It can provide additional transportation resources after disturbances, which enhances the ability of resistance of the overall network. Furthermore, different transportation modes offer functional complementarity. Under severe disturbances, affected passengers can switch to alternative modes to complete their journeys. Therefore, it is essential to consider both the functional interdependence between modes and the intermodal spatial interdependence when analyzing disturbance propagation in multimodal transportation networks.

In multimodal transportation networks, node failures necessitate route replanning by affected passengers. These routes often span multiple transportation modes, where intermodal coupling edges serve as critical connectors between modes. Additionally, betweenness centrality, derived from the shortest paths between nodes, is used to quantify functional interdependence within a multimodal network. Building on this, this study integrates edge betweenness centrality and passenger flow to characterize the functional role of intermodal coupling edges. This metric captures the temporal changes in the betweenness centrality of intermodal coupling edges within the remaining operational network at each time step following a disturbance. This analysis facilitates the evaluation of functional interdependence among transportation modes. To quantitatively assess the impact of intermodal coupling dynamics on network resilience, we construct the following formula:

$$FI_i^t = \frac{1}{2}\left(\frac{\sum_{j=1}^{N_C} BC_{ij}^t \cdot \delta_{ij}}{\sum_{i=1}^{N_C}\sum_{j=1}^{N_C} BC_{ij}^t} + \frac{\sum_{j=1}^{N_C} f_{ij}^t \cdot \delta_{ij}}{\sum_{i=1}^{N_C}\sum_{j=1}^{N_C} f_{ij}^t}\right) \tag{3}$$

$$BC_{ij}^t = \sum_{o,d\in N, o\neq d} \frac{p_{od}^{t,ij}}{p_{od}} \tag{4}$$

(3) Integrated interdependence factor

In urban multimodal transportation networks, the interaction between spatial and functional interdependence dimensions jointly determines the coupling dynamics across transportation modes. This study integrates these two dimensions to develop a comprehensive coupling factor that characterizes the intermodal coupling effects within multimodal transportation networks. The formula is as follows:

$$IC_i = (1-\zeta)\cdot SI_i + \zeta \cdot FI_i \tag{5}$$

where $\zeta \in [0, 1]$. This study sets balanced spatial and functional interdependence factors, $\zeta = 0.5$.

3.3. Multimodal network disturbance propagation model

In transportation networks, node failures disturb operations, and passenger rerouting leads to the propagation of these disturbances across the network. Cascading failures arise when disturbances originating from the initially failed nodes propagate to adjacent nodes and trigger further failures. This process continues until the network either stabilizes or experiences complete collapse. In multimodal transportation networks, disturbances caused by node failures propagate not only within individual modes but also across interconnected modes through intermodal couplings, as depicted in Fig. 2. However, most existing disturbance propagation models are designed for monomodal networks and primarily focus on intramodal propagation, which limits their effectiveness in simulating the cascading failures in multimodal networks. To address this limitation, this study proposes an improved CML model to analyze cascading failures in multimodal transportation networks. The improved model incorporates both intramodal and intermodal coupling relationships, which integrate spatial and functional interdependence factors to more accurately capture disturbance propagation.

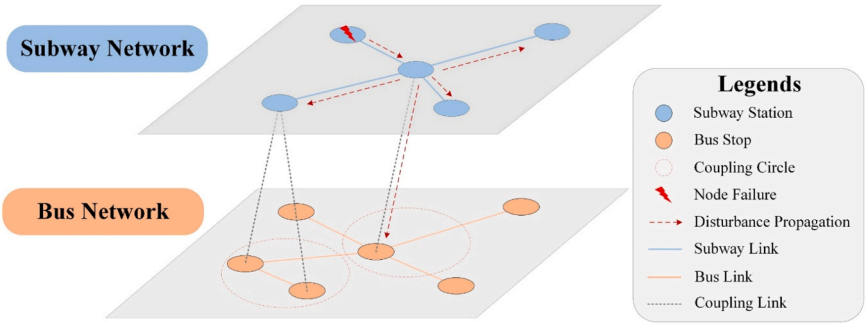


Fig. 2. Disturbance propagation process in multimodal transportation networks.

Furthermore, discrimination parameters for demand-type variations and intermodal coupling strength are introduced to enhance the generalizability of the model across diverse network configurations.

3.3.1. Improved CML model

In multimodal transportation networks, cascading failures propagate in two primary mechanisms: intramodal and intermodal propagation. Intramodal propagation occurs when a node failure within a single mode (e.g., subway or bus network) causes disruptions confined to the originating mode. In this case, passengers continue using the original transportation mode, leading to disturbance propagation within the same network. Intermodal propagation, on the other hand, arises when a node failure compels passengers to switch transportation modes, which enables disturbances to spread from one mode (e.g., subway) to another (e.g., bus). This results in cross-network cascading effects. Consequently, a holistic cascading failure model for multimodal transportation networks must account for both propagation mechanisms simultaneously.

Coupled map lattices serve as a fundamental framework for modeling cascading failures in transportation networks. This study proposes an improved CML-based model to analyze cascading failures in multimodal transportation networks, as illustrated in Fig. 3. The relevant algorithm process is illustrated in the Algorithm. The proposed refinements include three key advancements: 1) The measurement of inter-node relationships is enhanced. For intramodal disturbance propagation, network weights are determined by integrating network topology and monomodal travel demand. For intermodal disturbance propagation, intermodal coupling edges are defined using the maximum coupling radius distance and multimodal travel demand. Both structural and demand-based factors are incorporated to quantify coupling weights. 2) The characterization of intermodal coupling effects is improved. A comprehensive coupling factor, incorporating spatial proximity and functional interdependencies, is integrated to represent intermodal interactions. 3) The model generalizability is increased. Discrimination parameters for coupling relationship and coupling strength are introduced, enabling the model to analyze cascading failures in both monomodal and multimodal transportation networks.

(1) Initial node state

In the study of logical chaotic systems, accurately determining the initial state of a system is crucial. This is particularly important in analyses of cascading failures within multimodal transportation networks, where accurate identification of initial node states is fundamental for modeling subsequent state evolution. Current research often employs a random assignment method to allocate initial values between 0 and 1 across network nodes [6,39]. However, such an approach inadequately reflects the operational realities of transportation nodes. To address this limitation, this study uses node saturation, derived from empirical transit transaction data, as the initial states of the network. The calculation formula is as follows:

$$x_i^{initial} = \frac{q_i}{C_i} \tag{6}$$

(2) Node state evolution function

In the CML model, the state evolution of a node at time $t+1$ primarily depends on its current state and the states of its adjacent nodes at time t. Recent studies have aimed to enhance the state evolution function by incorporating travel demand for a more accurate characterization of network cascading processes. However, existing studies predominantly focus on monomodal transportation networks, which neglect the interplay of travel demand across multiple modes. To address this gap, this study integrates multimodal

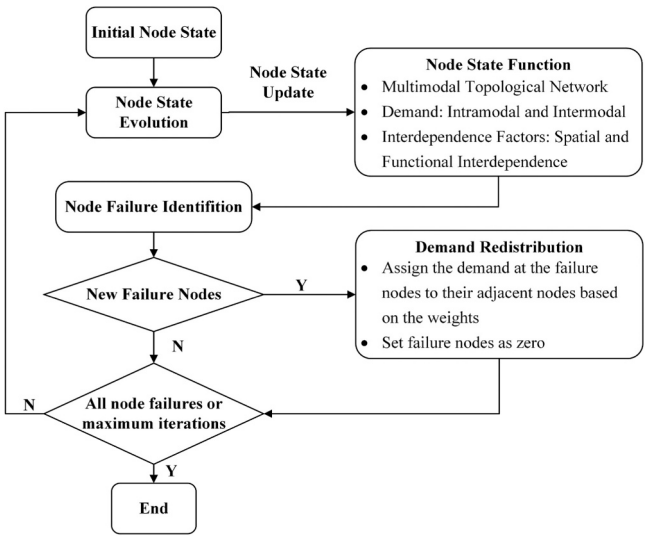


Fig. 3. Schematic diagram of the improved CML model.

transit transaction data, accounting for both the topological structure and travel demand of multimodal transportation networks to determine network weights. This approach quantifies the influence of adjacent nodes, which include intramodal connections and intermodal coupling nodes. Furthermore, this study introduces a comprehensive coupling factor into the CML model to capture intermodal interactions. The resulting state evolution function for network nodes is formulated as follows:

$$x_i(t+1) = \left| (1-\varepsilon)\cdot f(x_i(t)) + \varepsilon \cdot \sum_{j=1, j\neq i}^{N} \frac{\tau_{ij}\cdot W_{ij}\cdot f(x_j(t))}{S_i(t)} \right| + \gamma \cdot \frac{\sum_{j=1}^{N_C} IC_{ij}}{\sum_{i=1}^{N}\sum_{j=1}^{N_C} IC_{ij}} \tag{7}$$

$$W_{ij} = (1-\alpha)\cdot w_{ij}^{T} + \alpha \cdot w_{ij}^{D} \tag{8}$$

$$S_i(t) = \sum_{j=1, j\neq i}^{N} W_{ij} \tag{9}$$

$$w_{ij}^{T} = \frac{a_{ij}}{\sum_{j\in N} a_{ij}} \tag{10}$$

$$w_{ij}^{D} = \frac{f_{ij}}{\sum_{i\in N, j\in N, i\neq j} f_{ij}} \tag{11}$$

$$f_{ij} = \sum_{o,d\in N} q_{od}\cdot \rho_{ij} \tag{12}$$

$$f(x) = 4x(1-x) \tag{13}$$

3.3.2. Multimodal network node failure model

In a multimodal transportation network, the node state under stable conditions generally falls within the range (0, 1). At this stage, the evolution of the network state among nodes can be characterized by Eq. (7). However, if node i experiences an external disturbance (R) at time step $t+1$, this disturbance is incorporated as an additional factor into its state evolution function. For instance, during morning peak hours, excessive passenger flow accumulation at node i may lead to its overload-induced functional failure, i.e., $x_i(t+1) > 1$. Under these conditions, the state evolution function for the disturbed node can be expressed as:

$$x_i(t+1) = \left| (1-\varepsilon)\cdot f(x_i(t)) + \varepsilon \cdot \sum_{j=1, j\neq i}^{N} \frac{\tau_{ij}\cdot W_{ij}\cdot f(x_j(t))}{S_i(t)} \right| + \gamma \cdot \frac{\sum_{j=1}^{N_C} IC_{ij}}{\sum_{i=1}^{N}\sum_{j=1}^{N_C} IC_{ij}} + R \tag{14}$$

3.3.3. Demand redistribution

In multimodal transportation networks, the failure of node i may cause its complete functional disruption. The adjacent nodes are initially affected during this process. As a result, passenger flow previously routed through node i is redistributed to adjacent nodes, as

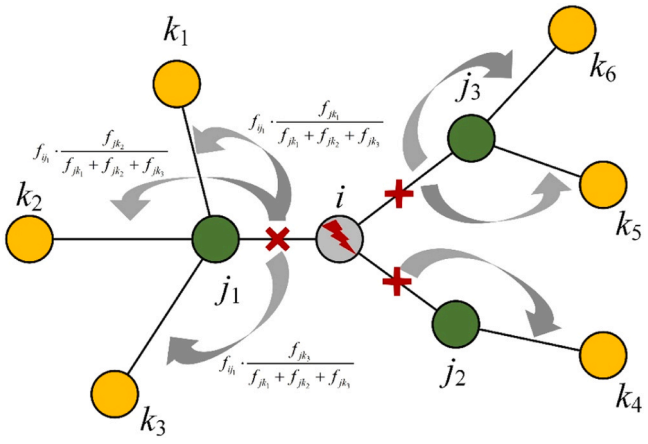


Fig. 4. Passenger flow distribution after node failure.

illustrated in Fig. 4. The proportion of this redistributed flow is determined by the passenger flow ratios on the remaining edges connecting the adjacent nodes. Given that multimodal transportation networks include both intramodal and intermodal connections, this study utilizes smart card transaction data from buses and subways to identify monomodal and transfer travel demands, thereby quantifying travel demand for both intramodal and intermodal edges. This approach ensures alignment with passenger flow characteristics in functional interdependence. From a behavioral perspective, passengers encountering a failed node typically opt for the nearest accessible transit station (e.g., subway station or bus stop) to continue their journey with minimal delay. This redistribution mechanism closely reflects real-world passenger behavior patterns. For instance, consider the failure of node i at time step t. The flow between node i and node j_1 is proportionally redistributed across edges connecting j_1 to nodes k_1, k_2, and k_3, based on the flow proportion of each edge relative to the total flow across all three edges. The corresponding calculation formula is as follows:

$$f_{jk}(t+1) = f_{jk}(t) + \Delta f_{jk} \tag{15}$$

$$\Delta f_{jk} = f_{ij} \cdot \frac{f_{jk}}{\sum_{k \in N_{cj}, k \neq i} f_{jk}} \tag{16}$$

Algorithm. The process of improved CML and demand redistribution

Input: the failure node *failedNode*; multimodal adjacent metrix *multimodalNetMat*; multimodal distance metrix *multimodalDisMat*; multimodal passenger flow metrix *multimodalFlow*; maximum iteration *maxIteration*; external disturbance R; new failure node set *newFailedNode*;

1. Initialize *newFailedNode* ← Φ;
2. Calculate the initial state of each node;
3. Give R to *failedNode* and update the node state according to Eq. (14);
4. Set $n \leftarrow 1$;
5. **Repeat** the following steps:
6. Determine the newly failed nodes and set *newFailedNode* ← the set of newly failed nodes;
7. **for** $node_i$ in *newFailedNode* **do**
8. Calculate the adjacent nodes $Node_j$ of $node_i$;
9. **for** $node_j$ in $Node_j$ **do**
10. Determine the flow f_{ij} from $node_i$ to $node_j$;
11. Check other edges connected to $node_j$; determine the intermodal and intramodal passenger flow in these edges, and calculate the proportions of each edge and the redistributed passengers as Eq. (16);
12. **end for**
13. Disconnect the edges with $node_i$ and update *multimodalNetMat*, *multimodalDisMat*, and *multimodalFlow*;
14. **end for**
15. Calculate the integrated interdependence factor *IC* of all the nodes according to *multimodalNetMat, multimodalDisMat*;
16. Update the node state according to Eq. (7) and set the node state in *newFailedNode* as 0;
17. Set $n \leftarrow n$+1;
18. **Until** $n >$ *maxIteration* or all the nodes are failed

3.4. Network performance indicators

In transportation network resilience studies, topology-based methods are widely employed to quantify and evaluate network performance. These approaches define performance metrics by examining alterations in node or edge states under network failure scenarios. Given that transit stations or stops represent the origins of passengers, the extent of node damage effectively reflects the service capacity of the system. The CML model proposed in this study captures cascading failures by modeling the dynamic evolution of node states. To assess the impact of cascading failures on network performance, this study employs three metrics: the failed node ratio, the newly failed nodes ratio at each time step, and the maximum connected subgraph. These metrics evaluate changes in disturbance propagation scale, network connectivity, and robustness under cascading failures.

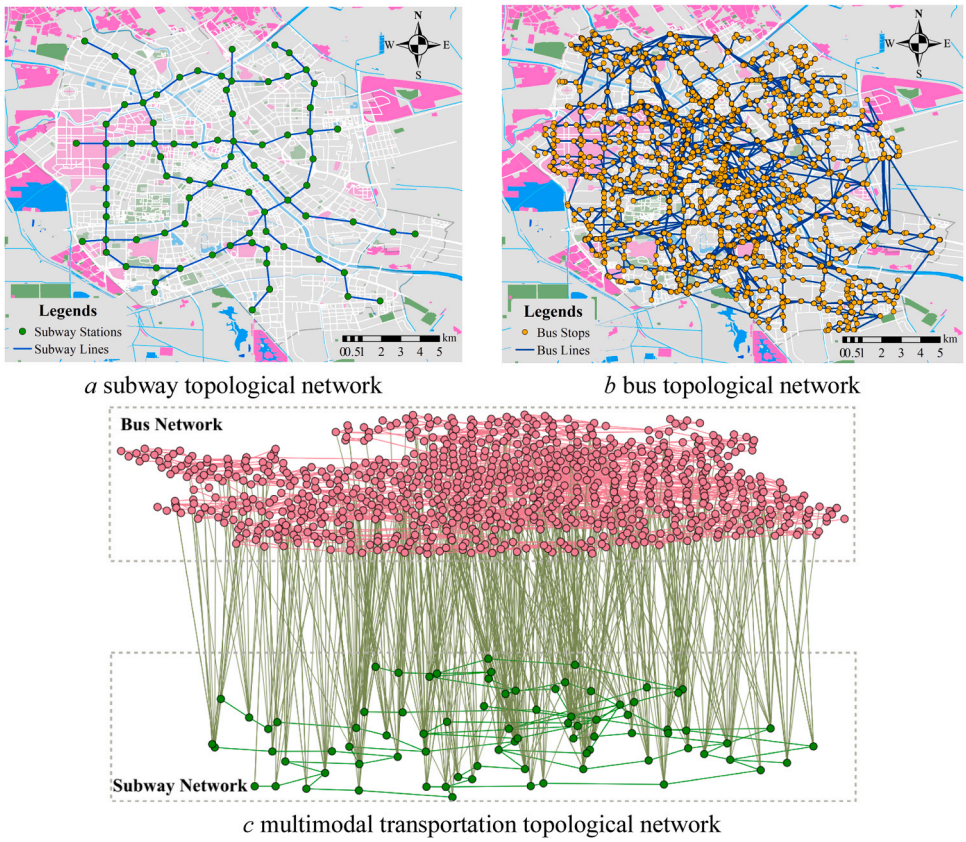
(1) Disturbance propagation scale

In multimodal transportation networks, severe disturbances can trigger cascading failures among nodes, leading to the progressive failure of additional nodes over time. This study quantifies the propagation extent of disturbances by analyzing the temporal evolution of the proportion of failed nodes relative to the total number of nodes in the network. The corresponding calculation formula is as follows:

$$c(t) = \frac{|N_d(t)|}{|N|} \tag{17}$$

(2) Network connectivity

Network connectivity is a critical metric for assessing a transportation network's ability to preserve topological connectedness, functional availability, and service accessibility during cascading failures. It provides a quantitative measure of the robustness of the network under such disturbances. In multimodal transportation network robustness research, the size ratio of the maximum connected subgraph (*MCS*) is widely used to evaluate network connectivity [35]. This indicator quantifies the changes in connectivity by calculating the ratio of nodes in the maximum connected subgraph relative to the total number of nodes after disturbances. This study



a subway topological network

b bus topological network

c multimodal transportation topological network

Fig. 5. Multimodal transportation topological network.

uses *MCS* at each time step to evaluate network robustness, which is formulated as follows:

$$MCS(t) = \frac{|N_{\max}(t)|}{|N|} \tag{18}$$

(3) Newly failed node ratio at each time step.

During a cascading failure process, the number of newly failed nodes may vary across time steps. In multimodal transportation networks, disturbance propagation occurs both within and between modes. By classifying failed nodes and conducting a statistical analysis, critical nodes in the disturbance propagation can be identified. Accordingly, this study proposes a network performance metric that incorporates newly failed nodes over time. This indicator can capture the temporal scale of disturbance propagation and identify key nodes in disturbance propagation, thereby enabling effective intervention in the cascading process, which is of significant importance. The specific formulations are given below:

$$p(t) = \frac{|N_d^{new}(t)|}{|N|} \tag{19}$$

$$N_d^{new}(t) = N_d^{M1}(t) \cup N_d^{M2}(t) \tag{20}$$

4. Case study

4.1. Multimodal transportation network construction

This study examines the bus and subway networks in six central districts in Tianjin, China, as a case study. By integrating the coupling distance threshold and multimodal travel demand, we establish intermodal coupling relationships to construct a "bus-subway" multimodal transportation network model, which consists of 940 nodes and 2063 edges, as illustrated in Fig. 5.

4.2. Data processing

This study analyzes travel demand using transit smart card data (including subway and bus systems) from Tianjin, China, on June 10, 2019. The raw data were first preprocessed to eliminate invalid records. For the bus system, which operates on a single-trip fare model, boarding stops were identified by integrating smart card data with GPS positioning information. Based on the spatiotemporal trajectory characteristics of passengers in the bus and subway systems, bus travel chains [48] and "bus-subway" intermodal travel chains were constructed, enabling the inference of bus alighting stops. For further analysis, travel records were filtered to include only trips with origins and destinations within the six central districts during the morning peak (07:00–08:00), evening peak (17:00–18:00), and the off-peak (13:00–14:00) periods. Travel demand for these three periods was then determined, including both monomodal and multimodal travel. Finally, the extracted travel demand data were integrated into a multimodal transportation network for subsequent analysis.

4.3. Numerical analysis

Due to the sensitivity of the CML model to parameter settings and initial states, this study investigates the threshold effects of key parameters in the improved CML model and examines the influence of different initial states during peak and off-peak hours. As disturbances in real-world scenarios may cause the failure of multiple nodes, and the transfer distance between transportation modes influences passenger transfer convenience, this study further analyzes the impact of coupling distance and the number of initially failed nodes on network robustness. Moreover, the improved CML model is applied to both monomodal and multimodal networks to verify its generalizability. Based on the statistics of failed nodes at each time step, critical nodes are identified. Finally, the improved CML model is compared with two traditional CML models to assess its enhanced capability in simulating cascading failures in multimodal transportation networks. In these experiments, initial failure nodes are selected based on travel demand or betweenness centrality. These experiments provide a comprehensive examination of cascading failure mechanisms in multimodal networks under various disturbance scenarios.

(1) Sensitivity analysis of key parameters

Table 2
Settings of parameter values in the sensitivity analysis of key parameters.

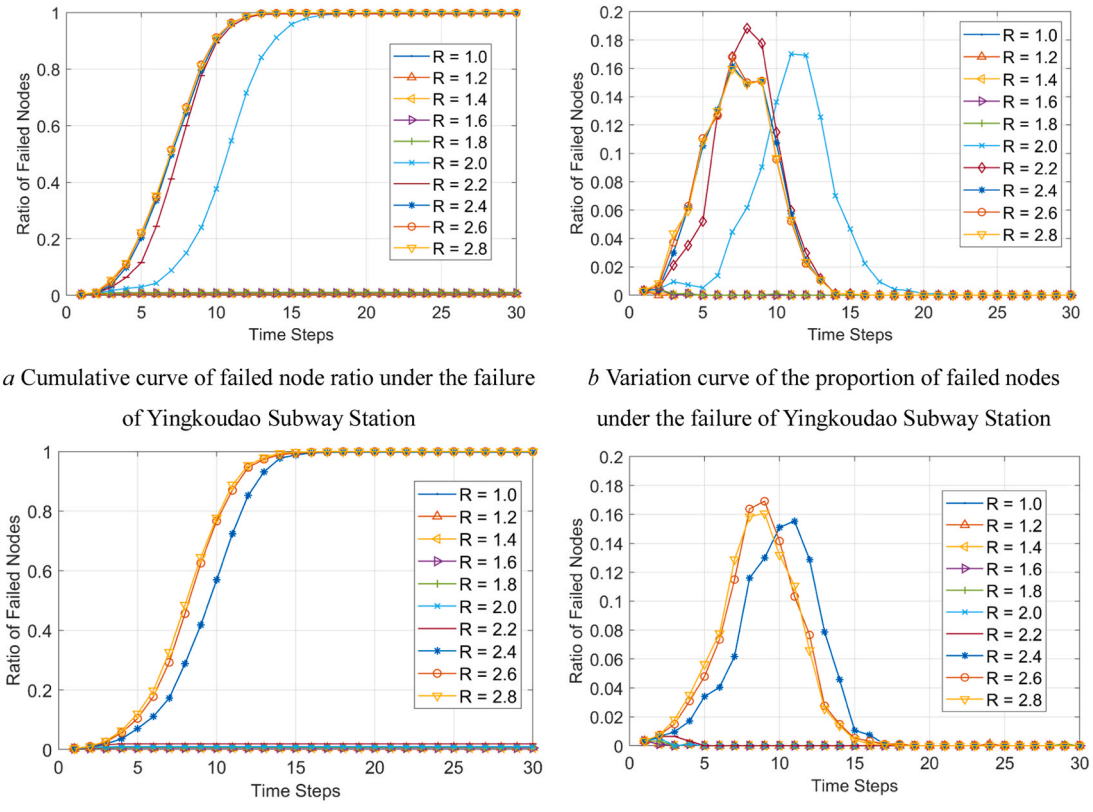
Key parameters in the model	Initial failed node	Parameters
Disturbance intensity (R)	Subway station (Yingkoudao) or Bus stop (Ertongcun)	$\varepsilon = 0.5$; $\gamma = 1$
Node correlation (ε)		$R = 2.4$; $\gamma = 1$
Intermodal coupling strength (γ)	Subway station (Yingkoudao)	$R = 0$; $\varepsilon = 0.5$

This experiment examines the threshold effect of three parameters, disturbance intensity (R) , node correlation (ε), and intermodal coupling strength (γ), in the improved CML model. The relevant settings of parameters are shown in Table 2.

The experimental results on the threshold effect of disturbance intensity (R) are presented in Fig. 6. At $R = 2.0$, the failure of the subway station with the highest travel demand (Yingkoudao Station) triggers cascading failures in the multimodal network. Similarly, at $R = 2.4$, the failure of the highest-demand bus stop (Ertongcun Stop) induces cascading failures. This comparison reveals that subway station failure propagates system-wide failures under lower disturbance intensity. This phenomenon arises from the significantly higher passenger flow in the subway network relative to the bus network. When a high-demand subway station fails, a large number of passengers are redistributed to adjacent edges, particularly connecting bus stops with lower capacities, which are more vulnerable to large-scale cascading failures. These findings underscore the critical role of the subway network as the backbone of urban transit systems. Furthermore, statistical analysis of temporal node failure proportions reveals that as R increases, the peak of the failure curve shifts to earlier time steps. However, once R exceeds a critical threshold (R=2.2 for subway station; R=2.6 for bus stop), the peak stabilizes within a fixed time window. This suggests that while heightened disturbance intensity accelerates the cascading failure propagation, exceeding a threshold does not significantly amplify failure rates.

The states of adjacent nodes play a crucial role in the interdependencies in the cascading failure process. This study examines the influence of node correlation (ε), with the experimental results presented in Fig. 7. The results indicate that when $\varepsilon \in (0.1 , 0.3)$, the correlation between nodes remains insufficient to propagate failures network-wide. A marked transition occurs at $\varepsilon = 0.4$ in the subway node failure scenario and $\varepsilon = 0.5$ in both the subway and bus node failure scenarios. This threshold-dependent behavior aligns with patterns observed in the analysis of disturbance intensity (R), which confirms the existence of a critical correlation threshold. To ensure the interdependence between nodes in multiple scenarios, a node correlation threshold of $\varepsilon = 0.5$ is adopted in subsequent experiments.

In the improved CML model, the integrated interdependence factor enhances the ability to capture the intermodal coupling effects of the model. The parameter γ determines the extent of this influence. To identify an appropriate value for γ, this experiment examines $\gamma \in (0.5, 2.5)$ to analyze how coupling strength affects node states. A critical threshold is observed at which the network shifts from a stable state to one with emergent node failures as γ increases. As shown in Fig. 8, no node failures occur within the range $\gamma \in (0.5, 1)$. For $\gamma \in (1.5, 2.5)$, intermodal coupling induces additional node failures, with the number of failed nodes increasing alongside the coupling strength. To mitigate such failures, $\gamma = 1$ is selected as the coupling strength for subsequent analysis.



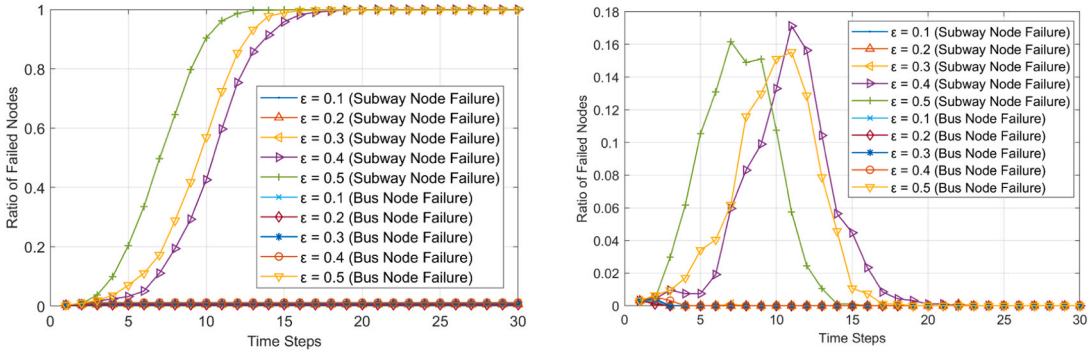
a Cumulative curve of failed node ratio under the failure of Yingkoudao Subway Station

b Variation curve of the proportion of failed nodes under the failure of Yingkoudao Subway Station

c Cumulative curve of failed node ratio under the failure of Ertongcun Bus Stop

d Variation curve of the proportion of failed nodes under the failure of Ertongcun Bus Stop

Fig. 6. Threshold analysis of disturbance intensity.



a Cumulative curve of failed node ratio under different node correlations

b Variation curve of the proportion of failed nodes under different node correlations

Fig. 7. Sensitivity of node correlation.

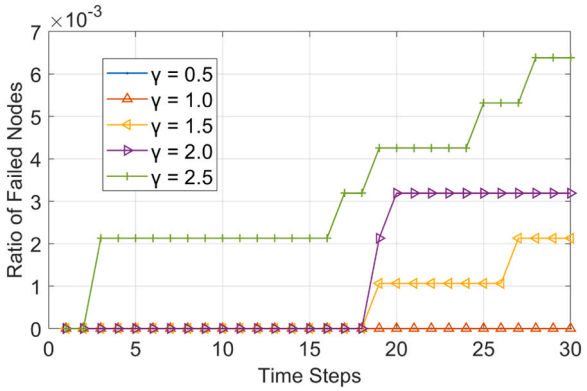


Fig. 8. The ratio of emergent failed nodes under different intermodal coupling strengths.

(2) Cascading failures during peak and off-peak hours

Through quantification of node saturation levels, the analysis investigates cascading failure processes arising from temporally differentiated node failures. Statistical analysis of daily travel patterns reveals a pronounced bimodal distribution, with a particularly prominent morning peak, as shown in Fig. 9. Consequently, travel demand from the morning peak (7:00–8:00), the evening peak (17:00–18:00), and the off-peak period (13:00–14:00) is used as the initial node-state data. The initial failure node is Yingkoudao Subway Station, and the parameters in this experiment are as follows: $\varepsilon = 0.5$, $\gamma = 1$. Results are presented in Fig. 10.

Experimental results demonstrate that cascading failures require a minimum disturbance intensity of $R = 2.0$ during the morning

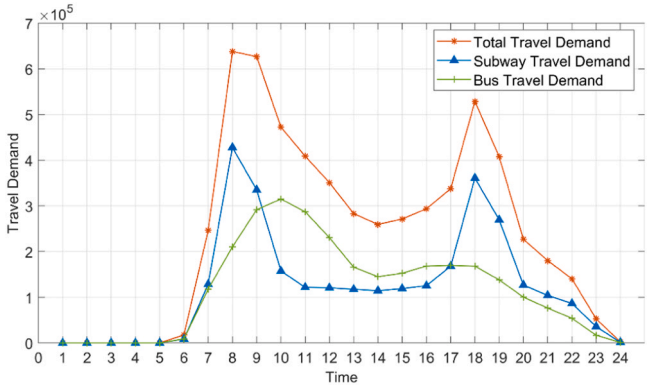
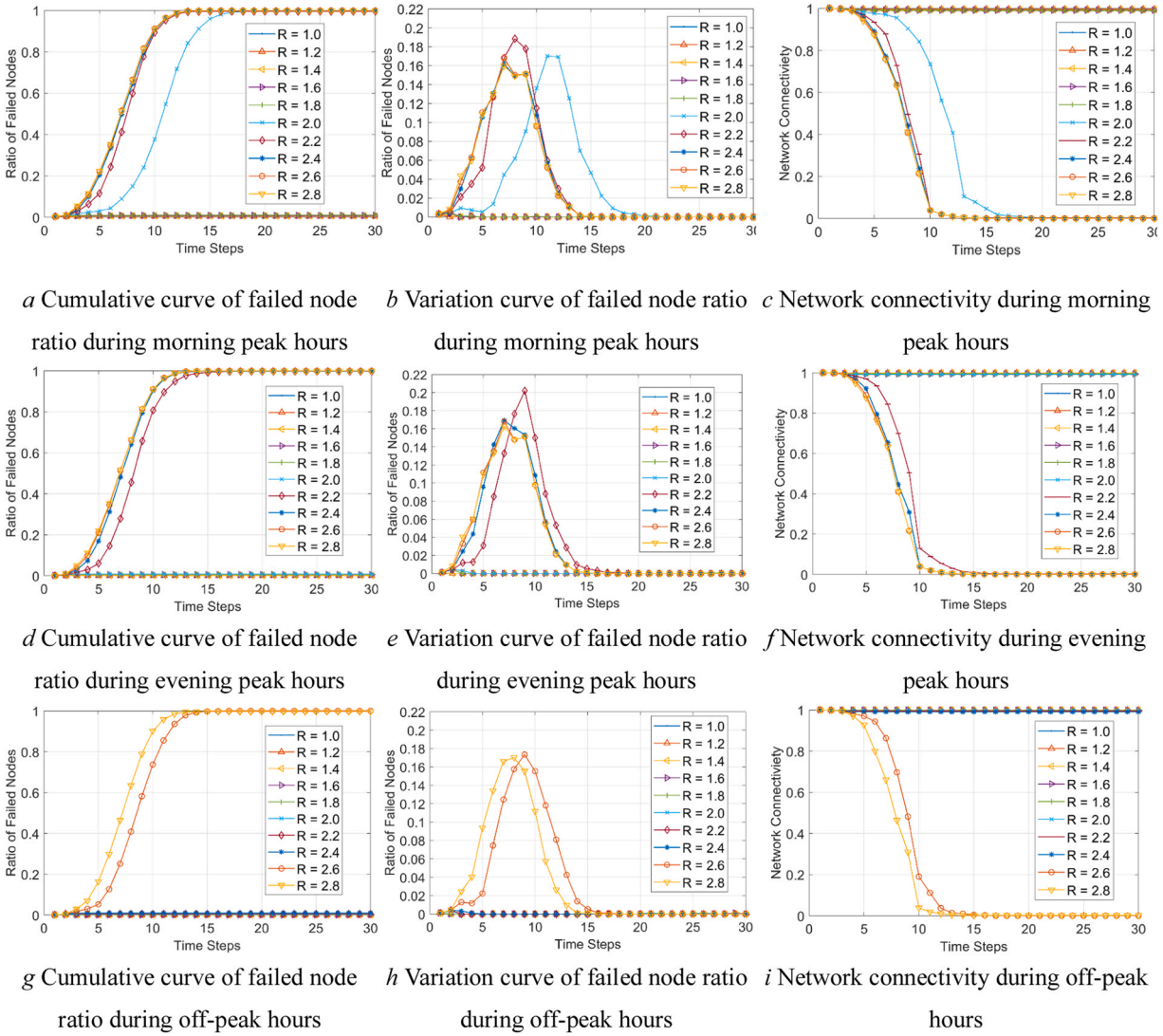


Fig. 9. The variation curve of daily travel demand.



a Cumulative curve of failed node ratio during morning peak hours

b Variation curve of failed node ratio during morning peak hours

c Network connectivity during morning peak hours

d Cumulative curve of failed node ratio during evening peak hours

e Variation curve of failed node ratio during evening peak hours

f Network connectivity during evening peak hours

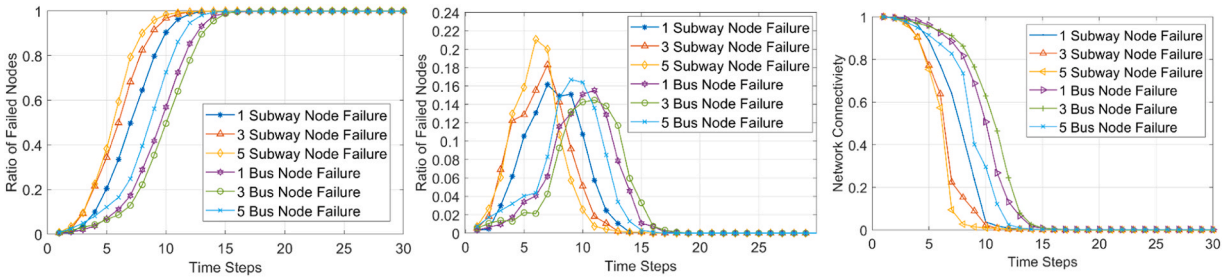
g Cumulative curve of failed node ratio during off-peak hours

h Variation curve of failed node ratio during off-peak hours

i Network connectivity during off-peak hours

Fig. 10. Network robustness under cascading failure during peak and off-peak hours.

peak, $R = 2.2$ during the evening peak, and a higher threshold of $R = 2.6$ during off-peak hours. These findings indicate an inverse relationship between initial node saturation and network robustness in topologically equivalent networks: higher initial node saturation reduces the resistance ability of the network to disturbances, while lower saturation increases residual load capacity, thereby



a Cumulative curve of failed node ratio

b Variation curve of failed node ratio

c Network connectivity

Fig. 11. The impact of the number of initial failure nodes on network robustness.

enhancing network robustness.

(3) The impact of multi-point failures in multimodal networks

To simulate the cascading failure in multimodal transportation networks under multi-node disturbances, this study identifies the top 1, 3, and 5 stations or stops with the highest morning peak-hour travel demand in both bus and subway networks as the initial failure nodes. The parameters in this experiment are as follows: $R = 2.4$, $\varepsilon = 0.5$, and $\gamma = 1$. Results are presented in Fig. 11. The results reveal that an increase in the number of initial failure nodes accelerates the network-wide node failure. Subway station failures lead to a decrease in network connectivity and full network collapse more quickly than equivalent failures at bus stops. Temporal analysis of the proportion of emergent failed nodes shows that subway station failures rapidly induce large-scale network failures, whereas bus stop failures require an accumulation of failed nodes exceeding a critical threshold. This phenomenon is primarily due to the broader propagation of disturbances in subway networks propagate more extensively when integrated with bus networks, since a single subway station often interconnects to multiple bus stops.

(4) The impact of coupling distance on multimodal network cascading failures

To assess the impact of coupling distance on cascading failures in multimodal transportation networks, this study constructs multiple network models with varying coupling distances and analyzes their relationship with cascading failures. The rise of short-distance transportation modes, such as shared bikes, has led to increased transfer distances. This experiment examines the coupling distances ranging from 200 m to 1000 m. The parameters used are as follows: $R = 2.4$, $\varepsilon = 0.5$, and $\gamma = 1$. The results are illustrated in Fig. 12. Between steps 6 and 11, the cumulative curve of the failed node ratio exhibits discrepancies, with the failure rate escalating as coupling distances increase. However, the network connectivity does not improve with the increase of coupling distance. This finding indicates that during the peak phase of the cascading failures, networks with longer coupling distances experience a significantly higher number of failed nodes, thereby exacerbating the vulnerability of networks under severe disturbances. It is worth noting that the short-distance transportation mode is not considered in other experiments, where a coupling distance of 200 m is adopted to reflect passenger transfer convenience.

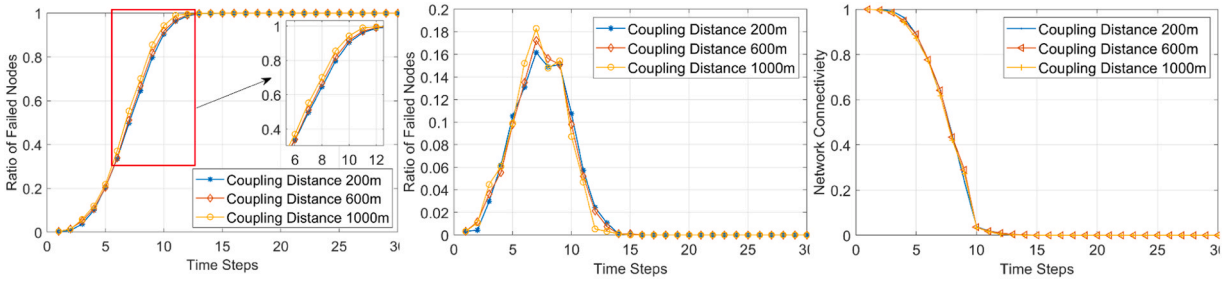
(5) Cascading failures in monomodal and multimodal networks

The improved CML model proposed in this study is applicable for analyzing cascading failures in both multimodal and monomodal transportation networks. To validate its generalizability, experiments are conducted on both network types. The initial failure node is the highest-demand station or stop. The parameters in this experiment are as follows: $\gamma = 0$ for the monomodal network, while $\gamma = 1$ for the multimodal network, and $\varepsilon = 0.5$. The results are presented in Fig. 13.

The results indicate that node failures initiate cascading failures at $R = 1.2$ in the subway network; At $R = 1.6$, these failures lead to complete network collapse, which indicates significant vulnerability. In the bus network, cascading failures result in network collapse at $R = 2.2$, reflecting its vulnerability. In the multimodal network, network-wide cascading failures occur at $R = 2.0$, positioning its vulnerability between the subway and bus networks. This pattern suggests that networks with higher saturation and simpler topologies exhibit reduced disturbance absorption capacity, whereas those with lower saturation and complex topologies exhibit greater robustness.

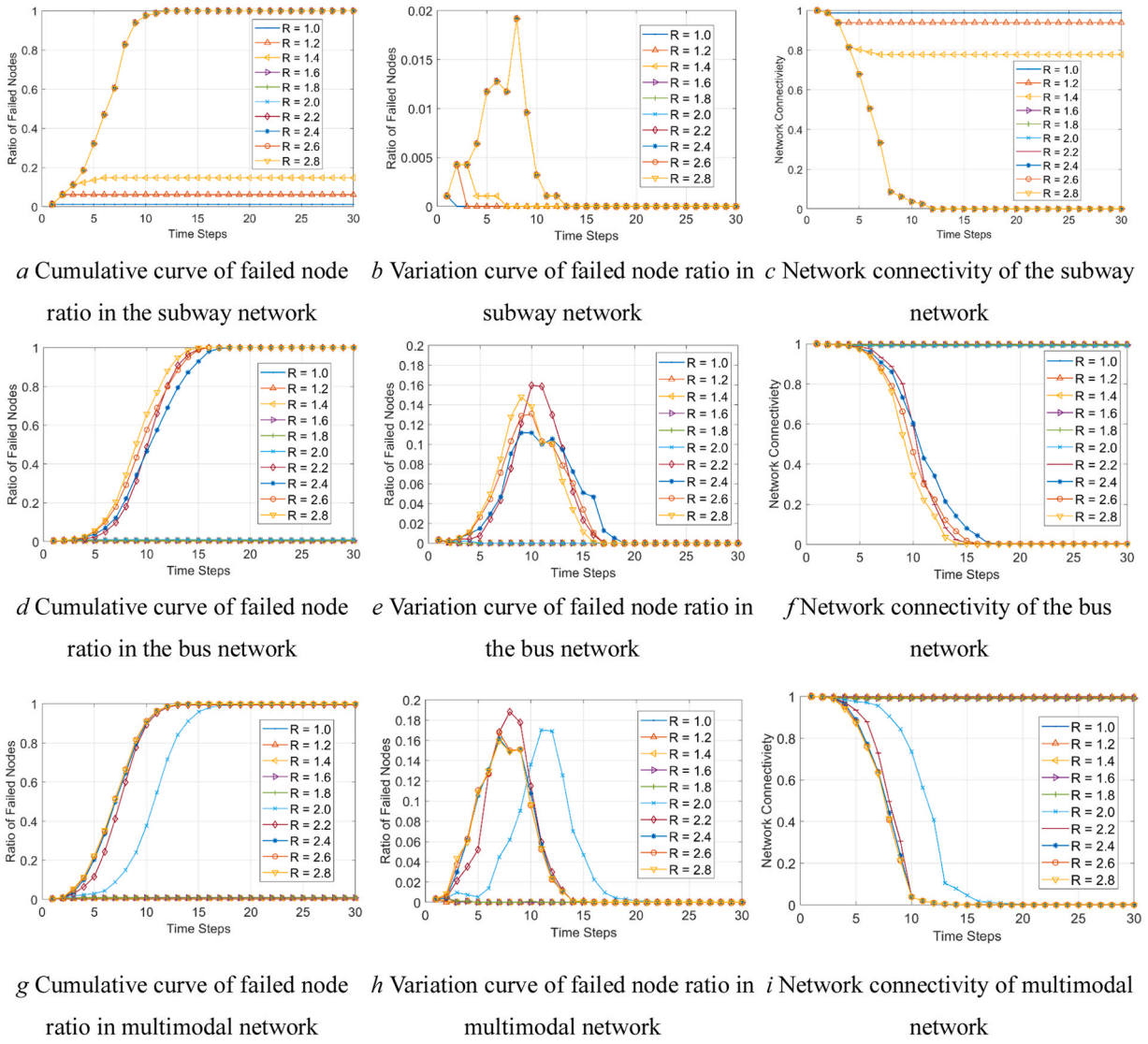
(6) Critical node identification

To mitigate cascading effects and minimize losses, it is critical to block disturbance propagation paths before large-scale node failures occur. In this experiment, the time interval of large-scale node failures is identified based on the variations in the proportion of newly failed nodes and changes in network performance. The preceding period is defined as the critical node identification interval.



a Cumulative curve of failed node ratio *b* Variation curve of failed node ratio *c* Network connectivity

Fig. 12. The impact of coupling distance on network robustness under cascading failures.



a Cumulative curve of failed node ratio in the subway network

b Variation curve of failed node ratio in subway network

c Network connectivity of the subway network

d Cumulative curve of failed node ratio in the bus network

e Variation curve of failed node ratio in the bus network

f Network connectivity of the bus network

g Cumulative curve of failed node ratio in multimodal network

h Variation curve of failed node ratio in multimodal network

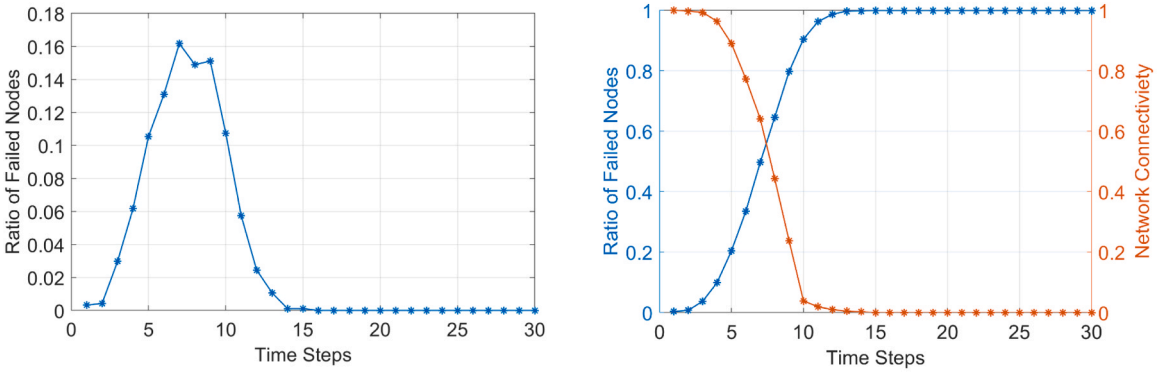
i Network connectivity of multimodal network

Fig. 13. Cascading failure in monomodal and multimodal transportation networks.

Critical nodes are identified by examining the number of newly failed nodes triggered by their neighbor nodes between consecutive time steps. As shown in Fig. 14, the proportion of failed nodes peaks between time steps 5 and 10, during which the failure ratio increases rapidly and network connectivity sharply declines, indicating significant network vulnerability. Therefore, time steps 1–5 are designated as the critical node identification interval. The corresponding statistical results are presented in Table 3. The results indicate that failures at subway stations are more likely to trigger intermodal disturbance propagation, whereas failures at bus stops tend to cause such propagation only after the fourth time step. Among the critical nodes identified, certain nodes appear with notably high frequency, such as the subway stations Yingkoudao (Station ID: 10), Xiaobailou (Station ID: 11), and the bus stops Dayingmen (Stop ID: 86), Nankai Sanmalu (Stop ID: 407), and Xiawafangditiezhan (Stop ID: 676). These nodes can thus be considered high-priority candidates for restoration. The simulation results provide valuable support for determining the prioritization of node recovery.

(7) Comparative analysis with other CML models

Existing CML models can be classified into two categories: those relying exclusively on network topology (Model I) [26] and those incorporating both network topology and demand characteristics (Model II) [6]. The formulas are shown in Eq. (21) and Eq. (22), respectively. This study examines the distinctions between an improved CML model and typical CML models by applying them to a multimodal transportation network and analyzing their respective failure propagation dynamics. The relevant parameters are shown in Table 4.



a failed node ratio at each time step

b Network connectivity and cumulative curve of failed node ratio

Fig. 14. Network robustness assessment and failed node ratio.

Table 3
Statistics of critical nodes at each time step.

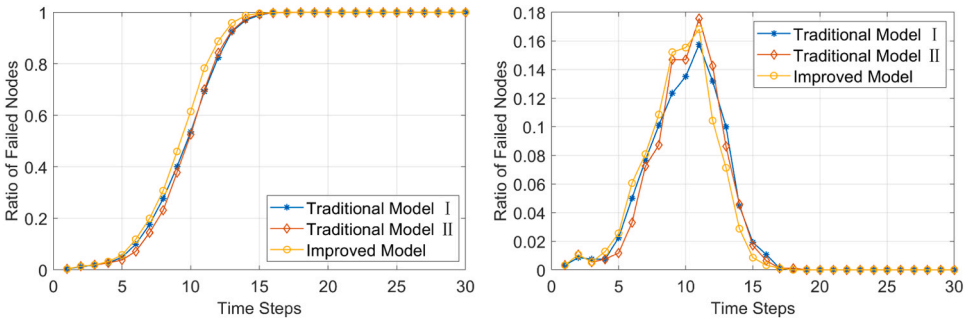
Time step	Initial Failure Node of Subway Station		Initial Failure Node of Bus Stop	
	Node ID (Intermodal)	Node ID (Intramodal)	Node ID (Intermodal)	Node ID (Intramodal)
1	10	10	-	127
2	9, 11, 36	9, 11, 35	-	86, 863, 938
3	8, 12, 37	443, 794, 86	-	314, 921, 779
4	50, 13, 49	407, 661, 676	167, 771, 913	913, 331, 634
5	51, 24, 33	655, 638, 793	11, 35, 10	676, 158, 407

Table 4
Settings of parameters in different CML models.

Models	Initial Failure Nodes	Parameters
Model I	The node with the highest betweenness centrality	$R = 2.4$; $\varepsilon = 0.5$
Model II		$R = 2.4$; $\varepsilon_1 = 0.25$; $\varepsilon_2 = 0.25$
The improved model		$R = 2.4$; $\varepsilon = 0.5$; $\gamma = 2$

$$x_i(t+1) = \left| (1-\varepsilon)\cdot f(x_i(t)) + \varepsilon \cdot \sum_{j=1, j\neq i}^{N} \frac{a_{ij}\cdot f(x_j(t))}{k_i} \right| \tag{21}$$

$$x_i(t+1) = \left| (1-\varepsilon_1-\varepsilon_2)\cdot f(x_i(t)) + \varepsilon_1 \cdot \sum_{j=1, j\neq i}^{N} \frac{a_{ij}\cdot f(x_j(t))}{k_i} + \varepsilon_2 \cdot \sum_{j=1, j\neq i}^{N} \frac{w_{ij}\cdot f(x_j(t))}{s_i} \right| \tag{22}$$



a Cumulative curve of failed node ratio

b Variation curve of failed node ratio

Fig. 15. Comparative analysis of the improved model and the traditional model.

The experimental results, illustrated in Fig. 15, indicate that all three models exhibit comparable trend patterns in the proportion of failed nodes across time steps, reaching peak value at $t = 11$. Notably, Model II, which incorporates travel demand, and the improved model exhibit higher peak node failure proportions compared to Model I during the cascading failure process, shown in Fig. 15*b*. Further analysis reveals that the improved model exhibits accelerated node failure rates relative to the other CML models. The statistics of the number of intermodal and intramodal failed nodes are illustrated in Table 5, and the spatial distribution of failed nodes at time steps 4, 6, and 8 is shown in Fig. 16. The experimental results show that Model II and the improved model outperform Model I in simulating intermodal perturbation propagation. Model I is more suitable for simulating cascading failure in a monomodal network. Compared with Model II, the improved model identifies more node failures due to intermodal propagation at time steps 4 and 6, with increases of 50.00 % and 46.67 %, respectively; Additionally, it captures more node failures due to intramodal propagation at time steps 4, 6, and 8, with increases of 84.62 %, 31.87 %, and 11.74 %, respectively. These results demonstrate that the improved CML model can more comprehensively simulate the cascading failure process of multimodal transportation networks. This enhanced capability provides essential support for suppressing disturbance propagation and formulating effective network recovery strategies, thereby reducing losses caused by disturbances.

5. Discussion and conclusions

5.1. Discussion

(1) The sensitivity analysis of model parameters and initial node states

The model parameters, disturbance intensity (R) and node correlation coefficient (ε), exhibit significant threshold effects. The magnitude of these thresholds is closely related to the transportation mode of the initially failed nodes. Compared to bus networks, subway networks typically carry higher passengers, which makes failures at subway stations more likely to trigger cascading failures across the entire network, even under lower levels of disturbance intensity and node correlation. Under severe disturbances, rapid estimation of failure propagation time, direction, and extent of failures becomes critical. Prioritizing the protection and restoration of the backbone infrastructure is essential to expedite overall network recovery.

Experimental results on the influence of the initial network state on cascading failures clarify the interdependence between transportation network dynamics and initial node states, and further corroborate the critical role of initial states in the evolution of chaotic systems. The initial network saturation levels during the morning peak, evening peak, and off-peak periods decrease sequentially, while the network's ability to withstand disturbances increases accordingly. This phenomenon reveals that greater residual capacity enhances the network's ability to absorb disturbances, thereby improving its robustness.

(2) Network robustness influencing factors

The number of initially failed nodes and the coupling distance significantly affect the robustness of multimodal transportation networks under cascading failures. The results indicate that increasing the number of initial failed nodes accelerates the rate of complete node failure. This is due, in part, to the larger number of initial disruptions and, more critically, to the fact that subway stations are often coupled with multiple bus stops. The failure of multiple subway stations leads to an exponential increase in the scale of the disturbance propagation. Consequently, the expansion of subway infrastructure necessitates explicit consideration of dynamic failure propagation during emergencies. Strategic increases in station density, alongside service coverage improvements, can enhance network robustness and mitigate the cross-system impacts of subway failures.

In addition, a larger coupling distance amplifies interactions between modes, hastening cascading failure propagation. The coupling distance analysis implies that stronger spatial interdependence between transportation modes does not inherently enhance network robustness. Conversely, the overly weak spatial interdependence may reduce passenger transfer efficiency and impact the passenger experience. Therefore, urban transportation infrastructure, such as subway stations, bus stops, and shared bicycle hubs, should be designed to balance infrastructure density to satisfy travel demand while effectively mitigating the vulnerability of the multimodal network.

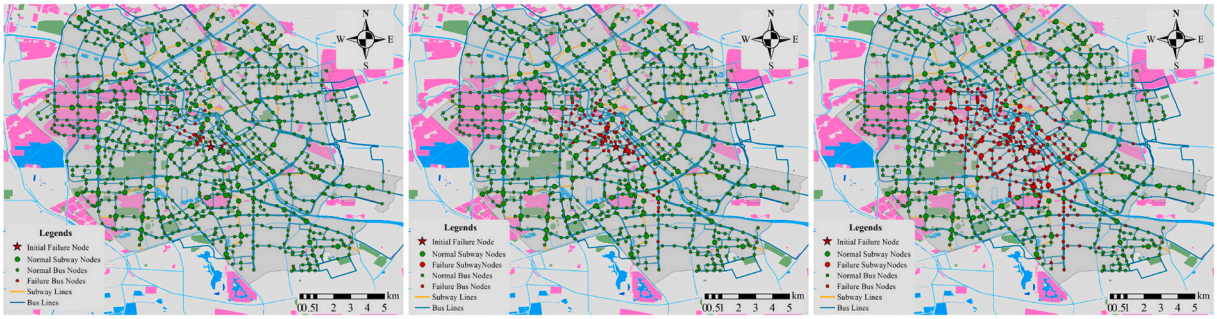
Experimental results on critical node identification show that subway station outages are more likely to cause widespread failures across the network, whereas disturbances originating from bus stop failures tend to remain confined to the bus network initially, with delayed propagation to the subway network. Accordingly, in the event of a subway station failure, it is essential to promptly allocate resources for repair or implement passenger evacuation strategies to contain the disturbance propagation before it leads to large-scale congestion.

(3) Effectiveness and generalizability of the improved CML model

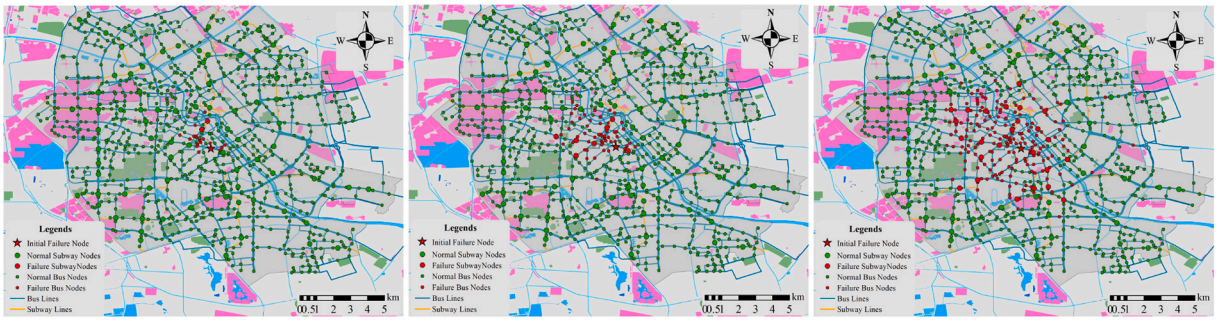
The improved CML model demonstrates robust generalizability and effectively captures the coupling effects in multimodal networks. By integrating demand-type discrimination parameters and coupling strength parameters, the model accommodates both monomodal and multimodal network analyses, thereby significantly broadening its applicability. The introduction of an integrated interdependence factor further refines its representation of intermodal failure propagation. Empirical results indicate that multimodal networks display intermediate characteristics, and their robustness to cascading failures falls between those of the two monomodal networks. Networks with high saturation and simple network structure are most vulnerable to failure, while low-saturation, complex

Table 5
Number of intramodal and intermodal failure nodes at different time steps.

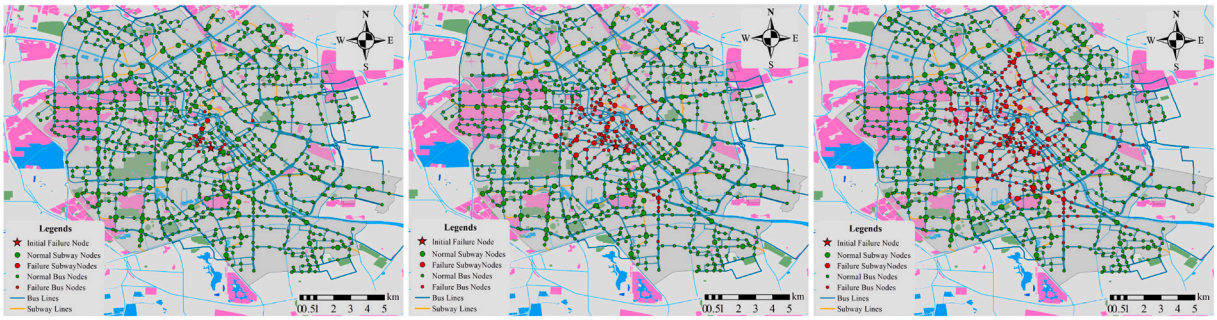
Time step	4		6		8	
	intermodal	intramodal	intermodal	intramodal	intermodal	intramodal
Model I	0	23	6	125	16	217
Model II	10	13	45	91	76	213
Improved Model	15	24	66	120	71	238



a Spatial distribution of failed nodes at time-step 4 for Model I *b* Spatial distribution of failed nodes at time-step 6 for Model I *c* Spatial distribution of failed nodes at time-step 8 for Model I



d Spatial distribution of failed nodes at time-step 4 for Model II *e* Spatial distribution of failed nodes at time-step 6 for Model II *f* Spatial distribution of failed nodes at time-step 8 for Model II



g Spatial distribution of failed nodes at time-step 4 for the improved Model *h* Spatial distribution of failed nodes at time-step 6 for the improved Model *i* Spatial distribution of failed nodes at time-step 8 for the improved Model

Fig. 16. Spatial distribution of failed nodes at time steps 4, 6, 8 in traditional and improved models.

networks are more resilient.

In the multimodal network construction, the intermodal coupling relationship is established by integrating spatial distance and multimodal travel demand. This approach establishes more accurate intermodal connections and more comprehensively captures the cascade failure process within the multimodal transportation network. Additionally, the independence factor is constructed by considering the heterogeneity of spatial distance, the node importance, and the dynamic demand, which further enhances the CML model's capacity to analyze cascade failures. The results demonstrate that the improved model, compared to other CML models, more effectively captures intermodal coupling effects during cascading failures in multimodal transportation networks.

5.2. Conclusions

This study proposes an improved CML model to systematically analyze cascading failure dynamics in urban multimodal transportation networks. To improve the accuracy of the CML model for multimodal network analysis, intramodal and intermodal travel demand are derived from smart card data, and a weighted multimodal transportation network is constructed by incorporating network topology and travel demand. Moreover, a comprehensive coupling factor that integrates spatial and functional interdependence is proposed, which is subsequently embedded into the CML model. Furthermore, the model incorporates discrimination parameters for the coupling relationship and coupling strength. Through parameter calibration, the improved CML model demonstrates generalizability across both monomodal and multimodal networks. Finally, a case study of bus-subway networks in six central districts in Tianjin, China, is conducted to validate the validity of the proposed model.

The study performs sensitivity analyses on model parameters and initial node states, and evaluates the impacts of the number of initial failure nodes and coupling distance on cascading failure propagation, thereby validating the effectiveness of the proposed model. Through systematic analysis, the key findings are summarized as follows: 1) The threshold effects of R and ε are influenced by the transportation mode of the initially failed nodes and initial node states. Failures in high passenger flow transportation networks, such as subway networks, are more likely to trigger network cascade failures. Greater residual capacity enhances the overall network robustness; 2) Establishing an appropriate coupling relationship between transportation modes is essential. Designing transfer stations that balance passenger transfer convenience with network interdependence can optimize transport functionality and reduce losses caused by cascading failures in the network. 3) The improved CML model can effectively capture intermodal coupling effects during cascading failures in multimodal transportation networks. It achieves a more balanced consideration of intramodal and intermodal node interactions, thereby providing a more precise representation of failure dynamics in such networks.

Building upon the aforementioned research findings, this study proposes the following strategies for urban transportation management to mitigate cascading failures in multimodal transportation networks effectively: 1) During infrastructure construction, optimal intermodal coupling distances should be identified for station or stop layouts to establish well-structured intermodal connections. This approach balances passenger transfer convenience while minimizing network vulnerability; 2) Strengthening system resilience requires reducing network saturation, particularly in high-capacity modes, through infrastructure expansion and increased node density; 3) Cascading failure models should be employed to simulate evolutionary processes during multi-node failures, offering actionable insights for rapid response.

The limitations of this study and potential directions for future research are as follows: 1) This study does not consider the recovery mechanisms after failure. In practice, failed nodes often resume functionality after repairs. Future work should integrate recovery dynamics to improve the realism of network evolution models. 2) Due to data limitations, the proposed model excluded factors such as the physical interpretation of iterative time steps, bus lines, and vehicle frequency. Future research may incorporate these elements into the modeling of network interdependence to further improve model accuracy. 3) Demand redistribution in the existing CML model relies primarily on network topology or edge weights, which constrain its ability to capture passengers' travel behavior during disturbances. Future research could integrate neural network approaches (e.g., PMR-GCN [49]) with empirical travel data collected during disturbances to better model passenger responses to station or stop failures. Incorporating these insights would improve demand redistribution accuracy.

CRediT authorship contribution statement

Biao Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Yanyan Chen:** Supervision. **Jianjun Wu:** Supervision. **Zhengbing He:** Supervision, Resources. **Der-Horng Lee:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the National Natural Science Foundation of China (No. U2469201), the 111 Project of China (No. B20071), and the Beijing Natural Science Foundation (No. L241079).

Data availability

The authors are unable or have chosen not to specify which data has been used.

References

[1] S. Pan, H. Yan, J. He, Z. He, Vulnerability and resilience of transportation systems: a recent literature review, Phys. A Stat. Mech. Appl. 581 (2021) 126235.
[2] D. Xue, L. Wu, T. Xu, C.L. Wu, Z. Wang, Z. He, Space weather effects on transportation systems: a review of current understanding and future outlook, Space Weather 22 (12) (2024) e2024SW004055.
[3] S. Pan, S. Ling, N. Jia, Y. Liu, Z. He, On the dynamic vulnerability of an urban rail transit system and the impact of human mobility, J. Transp. Geogr. 116 (2024) 103850.
[4] B.L. Dou, X.G. Wang, S.Y. Zhang, Robustness of networks against cascading failures, Phys. A Stat. Mech. Appl. 389 (11) (2010) 2310–2317.
[5] S.V. Buldyrev, R. Parshani, G. Paul, H.E. Stanley, S. Havlin, Catastrophic cascade of failures in interdependent networks, Nature 464 (7291) (2010) 1025–1028.
[6] A. Huang, H.M. Zhang, W. Guan, Y. Yang, G. Zong, Cascading failures in weighted complex networks of transit systems based on coupled map lattices, Math. Probl. Eng. 2015 (1) (2015) 940795.
[7] Q.C. Lu, J. Li, P.C. Xu, L. Zhang, X. Cui, Modeling cascading failures of urban rail transit network based on passenger spatiotemporal heterogeneity, Reliab. Eng. Syst. Saf. 242 (2024) 109726.
[8] M. Saberi, H. Hamedmoghadam, M. Ashfaq, S.A. Hosseini, Z. Gu, S. Shafiei, M.C. González, A simple contagion process describes spreading of traffic jams in urban networks, Nat. Commun. 11 (1) (2020) 1616.
[9] J. Huang, Q. Xu, Y. Yan, A model for airport congestion propagation in Chinese airline network. 2019 Chinese Automation Congress (CAC), IEEE, 2019, November, pp. 890–894.
[10] C. Li, X.M. Zhao, D.F. Xie, Z. He, C. Lu, Exploring the impact mechanism of communicating information on the perturbation propagation, Transportmet. A Transp. Sci. 20 (1) (2024) 2035013.
[11] J.J. Wu, Z.Y. Gao, H.J. Sun, H.J. Huang, Congestion in different topologies of traffic networks, Europhys. Lett. 74 (3) (2006) 560.
[12] Y. Shen, G. Song, H. Xu, Y. Xie, Model of node traffic recovery behavior and cascading congestion analysis in networks, Phys. A Stat. Mech. Appl. 545 (2020) 123422.
[13] K. Jin, W. Wang, X. Li, S. Chen, S. Qin, X. Hua, Cascading failure in urban rail transit network considering demand variation and time delay, Phys. A Stat. Mech. Appl. 630 (2023) 129290.
[14] J. Wu, Z.Y. Gao, H.J. Sun, Effects of the cascading failures on scale-free traffic networks, Phys. A Stat. Mech. Appl. 378 (2) (2007) 505–511.
[15] J.J. Wu, Z.Y. Gao, H.J. Sun, Optimal traffic networks topology: a complex networks perspective, Phys. A Stat. Mech. Appl. 387 (4) (2008) 1025–1032.
[16] D. Li, Y. Jiang, R. Kang, H. Shlomo, Spatial correlation analysis of cascading failures: congestions and blackouts, Sci. Rep. 4 (1) (2014) 5381.
[17] J. Zhao, D. Li, H. Sanhedrai, R. Cohen, S. Havlin, Spatio-temporal propagation of cascading overload failures in spatially embedded networks, Nat. Commun. 7 (1) (2016) 10094.
[18] P. Bak, C. Tang, K. Wiesenfeld, Self-organized criticality: an explanation of the 1/f noise, Phys. Rev. Lett. 59 (4) (1987) 381.
[19] I. Dobson, B.A. Carreras, V.E. Lynch, D.E. Newman, An initial model for complex dynamics in electric power system blackouts, hicss (2001, January).
[20] Y. Moreno, J.B. Gómez, A.F. Pacheco, Instability of scale-free networks under node-breaking avalanches, Europhys. Lett. 58 (4) (2002) 630.
[21] F. Wu, J. Yang, H. Jiang, X. Zhan, S. Liao, J. Xu, J. Liang, Cascading failure in coupled networks of transportation and power grid, Int. J. Electr. Power Energy Syst. 140 (2022) 108058.
[22] Z. Jin, D. Duan, N. Wang, Cascading failure of complex networks based on load redistribution and epidemic process, Phys. A Stat. Mech. Appl. 606 (2022) 128041.
[23] R. Yin, K. Zhang, X. Ma, Y. Wang, L. Li, Analysis of cascading failures caused by mobile overload attacks in scale-free networks, Phys. A Stat. Mech. Appl. 615 (2023) 128598.
[24] L. Sun, Y. Huang, Y. Chen, L. Yao, Vulnerability assessment of urban rail transit based on multi-static weighted method in Beijing, China, Transp. Res. Part A Policy Pract. 108 (2018) 12–24.
[25] Y. Shen, G. Ren, B. Ran, Cascading failure analysis and robustness optimization of metro networks based on coupled map lattices: a case study of Nanjing, China, Transportation 48 (2) (2021) 537–553.
[26] X. Xu, A. Huang, A. Shalaby, Q. Feng, M. Chen, G. Qi, Exploring cascading failure processes of interdependent multi-modal public transit networks, Phys. A Stat. Mech. Appl. 638 (2024) 129576.
[27] J. Duan, D. Li, H.J. Huang, Reliability of the traffic network against cascading failures with individuals acting independently or collectively, Transp. Res. Part C Emerg. Technol. 147 (2023) 104017.
[28] Y. Shen, H. Yang, G. Ren, B. Ran, Model cascading overload failure and dynamic vulnerability analysis of facility network of metro station, Reliab. Eng. Syst. Saf. 242 (2024) 109711.
[29] D.H. Kim, A.E. Motter, Resource allocation pattern in infrastructure networks, J. Phys. A Math. Theor. 41 (22) (2008) 224019.
[30] F. Ma, W. Shi, K.F. Yuen, Q. Sun, X. Xu, Y. Wang, Z. Wang, Exploring the robustness of public transportation for sustainable cities: a double-layered network perspective, J. Clean. Prod. 265 (2020) 121747.
[31] X. Guo, Q. Du, Y. Li, Y. Zhou, Y. Wang, Y. Huang, B. Martinez-Pastor, Cascading failure and recovery of metro–bus double-layer network considering recovery propagation, Transp. Res. Part D Transp. Environ. 122 (2023) 103861.
[32] X.F. Wang, J. Xu, Cascading failures in coupled map lattices, Phys. Rev. EStat. Nonlinear Soft Matter Phys. 70 (5) (2004) 056113.
[33] L. Zhang, M. Xu, S. Wang, Quantifying bus route service disruptions under interdependent cascading failures of a multimodal public transit system based on an improved coupled map lattice model, Reliab. Eng. Syst. Saf. 235 (2023) 109250.
[34] Q.C. Lu, L. Zhang, P.C. Xu, X. Cui, J. Li, Modeling network vulnerability of urban rail transit under cascading failures: A Coupled Map Lattices approach, Reliab. Eng. Syst. Saf. 221 (2022) 108320.
[35] C. Gao, Y. Fan, S. Jiang, Y. Deng, J. Liu, X. Li, Dynamic robustness analysis of a two-layer rail transit network model, IEEE Trans. Intell. Transp. Syst. 23 (7) (2021) 6509–6524.
[36] J. Li, Q.C. Lu, P.C. Xu, S. Wang, C. Xie, Cascading failures on multimodal public transportation networks: the role of station coupling strength, IEEE Trans. Intell. Transp. Syst. (2024).
[37] J. Zhang, Z. Wang, S. Wang, W. Shao, X. Zhao, W. Liu, Vulnerability assessments of weighted urban rail transit networks with integrated coupled map lattices, Reliab. Eng. Syst. Saf. 214 (2021) 107707.
[38] D.J. Sun, S. Guan, Measuring vulnerability of urban metro network from line operation perspective, Transp. Res. Part A Policy Pract. 94 (2016) 348–359.
[39] S. Wu, Y. Zhu, N. Li, Y. Wang, X. Wang, D.J. Sun, Urban rail transit system network reliability analysis based on a coupled map lattice model, J. Adv. Transp. 2021 (1) (2021) 5548956.
[40] Y. Zhao, L. Li, Z. Zhang, D. Sun, Performance evaluation for the expansion of multi-level rail transit network in Xi'an metropolitan area: empirical analysis on accessibility and resilience, Land 13 (10) (2024) 1682.
[41] X. Li, M. Teng, S. Jiang, Z. Han, C. Gao, V. Nekorkin, P. Radeva, A dynamic station-line centrality for identifying critical stations in bus-metro networks, Chaos Solitons Fractals 194 (2025) 116102.
[42] J. Sun, K. Bathgate, S. Pan, Z. Zhang, Network-based method for assessing multi-modal transportation network vulnerability to cascading failures, Sustain. Anal. Model. 4 (2024) 100034.

[43] L. Hong, Y. Yan, M. Ouyang, H. Tian, X. He, Vulnerability effects of passengers' intermodal transfer distance preference and subway expansion on complementary urban public transportation systems, Reliab. Eng. Syst. Saf. 158 (2017) 58–72.
[44] X.L. An, L. Zhang, J.G. Zhang, Research on urban public traffic network with multi-weights based on single bus transfer junction, Phys. A Stat. Mech. Appl. 436 (2015) 748–755.
[45] H. Zhang, C.X. Zhuge, X. Zhao, W.B. Song, Assessing transfer property and reliability of urban bus network based on complex network theory, Int. J. Mod. Phys. C. 29 (01) (2018) 1850004.
[46] R. Ding, N. Ujang, H.B. Hamid, M.S.A. Manan, R. Li, S.S.M. Albadareen, J. Wu, Application of complex networks theory in urban traffic network researches, Netw. Spat. Econ. 19 (2019) 1281–1317.
[47] B. Wu, A. Tang, J. Wu, Modeling cascading failures in interdependent infrastructures under terrorist attacks, Reliab. Eng. Syst. Saf. 147 (2016) 1–8.
[48] M. Jin, M. Wang, Y. Gong, Y. Liu, Spatio-temporally constrained origin–destination inferring using public transit fare card data, Phys. A Stat. Mech. Appl. 603 (2022) 127642.
[49] C. Gao, H. Liu, J. Huang, Z. Wang, X. Li, X. Li, Regularized spatial-temporal graph convolutional networks for metro passenger flow prediction, IEEE Trans. Intell. Transp. Syst. 25 (9) (2024) 11241–11255.